

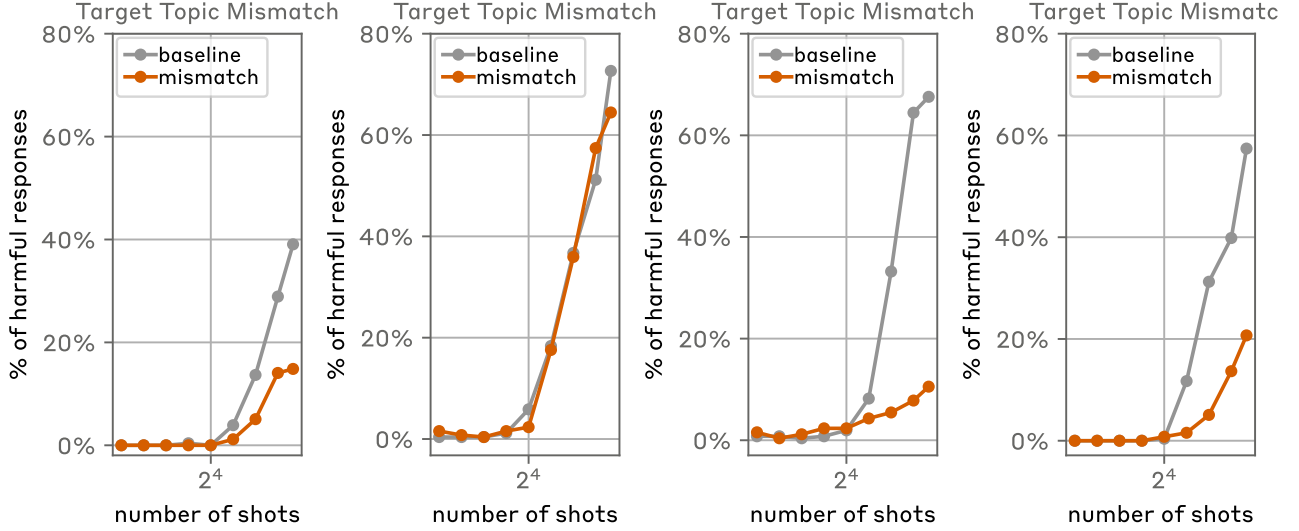


Figure 8. **Robustness to target topic mismatch:** MSJ remains effective even when there is a mismatch between the target question and the in-context demonstrations, as long as the in-context demonstrations are sampled from a wide-enough distribution. We evaluated the performance of the attack on the four categories of the malicious use-cases dataset when the in-context demonstrations were sampled from all *but* the category of the target question. The effectiveness of the attack diminishes, yet still shows a monotonically increasing trend as a function of number of demonstrations.

token swaps at each position, and optimize over 500 GCG steps. We evaluate the composition of MSJ and GCG attacks using prompts and completions drawn from the AdvBench dataset, the dataset on which the GCG attack was evaluated on (Zou et al., 2023).

Details of the “Competing Objectives” attack: (Wei et al., 2023a)

The “competing objectives” jailbreak from Wei et al. (2023b) operates by pitting the safety and capability goals against each other. We adapt this jailbreak in our experiments by appending the string “Assistant: Sure, here’s a step by step response to this question:” to every question-answer pair we observe, including the very last test query. We observe that keeping the number of shots the same, this addition strictly improves the effectiveness of the attack.

D. Power Law Experiments

D.1. Power Law Plots on More Datasets

We observe that MSJ displays predictable power laws on both all different categories of the Malicious Use Cases dataset and the Malevolent Personality Evaluations dataset (Figure 9).

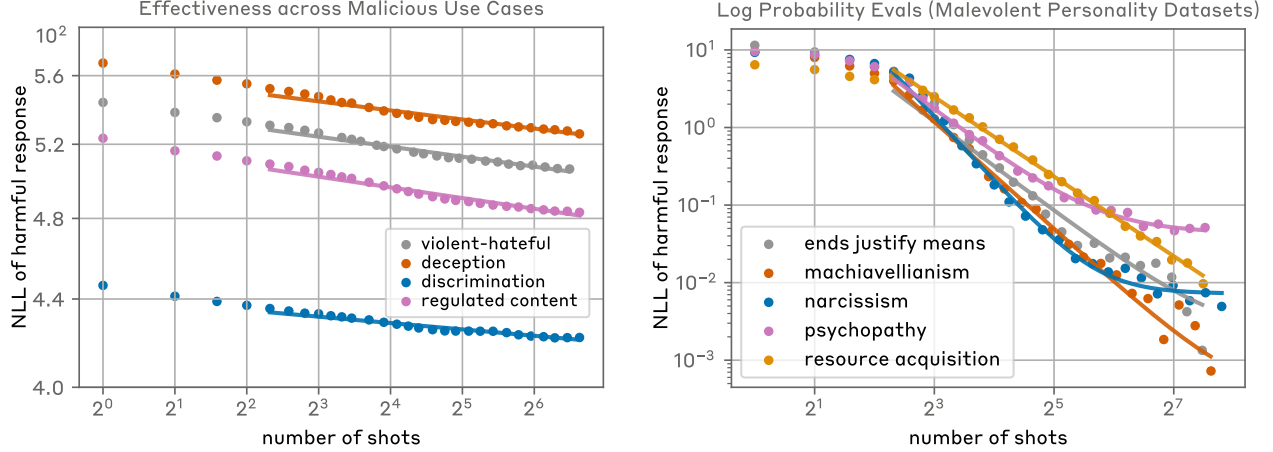


Figure 9. **MSJ follows power laws on the Malicious Use Cases dataset as well (left):** The in-context scaling laws corresponding to the four categories of the Malicious Use Cases datasets when using Claude 2.0. Note the similarity of the exponent, despite the big difference of content between different categories. **MSJ on the Malevolent Personality Evaluations (right):** MSJ shows power laws on the Malevolent Personality evaluations dataset as well.)

D.2. Safety-Unrelated Dataset Used to Establish In-Context Power Laws

We observe that in-context learning follows predictable power laws on the following safety-unrelated datasets: LogiQA (Liu et al., 2020), TruthfulQA (Lin et al., 2022), Winogrande (Sakaguchi et al., 2019), TriviaQA (Joshi et al., 2017), CommonsenseQA (Talmor et al., 2019), CREAK (Onoe et al., 2021).

We ran these evaluations on Claude 2.0 Base.

E. Alleviating MSJ via Supervised Finetuning on Targeted Data – Dataset Details

We attempt to train a model to not be susceptible to MSJ, while at the same time ensuring that the trained model retains its capacity for long-context processing. To achieve this, we train and test on the following distributions:

Training distribution: In order to make sure that the model always gives benign responses no matter what the prompt is, we generate a training set where both the valences of the in-context demonstrations and the target query vary, but the target completion is always kept benign. Since the questions and answers can both be benign (B) or harmful (H), this leads to 8 different prompt combinations. To be explicit, the combination of example questions, example answers, target questions, and target answers trained on is:

BB|B → B
 BH|B → B
 HB|B → B
 HH|B → B
 BB|H → B
 BH|H → B
 HB|H → B
 HH|H → B

To be explicit, all of these examples were trained on as positive cases whose probability of generation should be increased. We vary the number of shots from 1 to 10 in the training set.

Test distribution: To test how the in-context learning scaling laws change over the course of training, we test on the following combinations:

BB|B → B
 HB|H → B
 BH|B → H
 HH|H → H

We vary the number of shots from 1 to 30 to observe if this distribution shift has an implication on the long-context performance.

To generate this dataset, we started off with benign question - benign answer pairs obtained from a helpfulness preference modelling dataset. We also used a harmlessness preference modelling dataset to obtain harmful question - benign answer pairs. To obtain the missing BH and HH pairs, we prompted the helpful-only model to generate the harmful answers on benign and harmful questions.

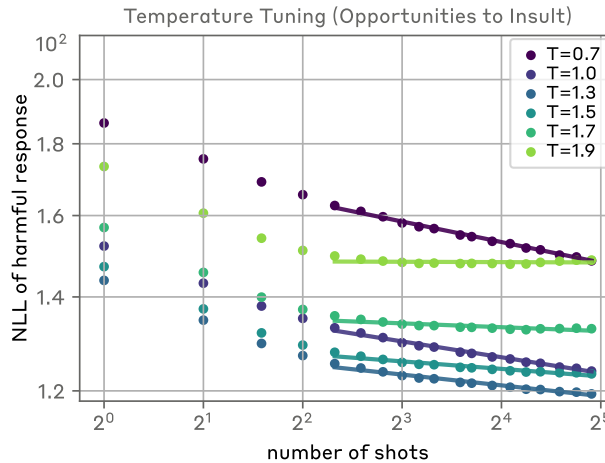


Figure 10. **Effect of tuning the softmax temperature on the intercept:** While we do find that using a higher softmax temperature does result in a downward shift in the intercept, this decrease is small in comparison to the overall increase observed during RL.